

MediConnect Healthcare Prior Authorization & Interoperability Platform

Functional Requirements

Purpose: Define the core system capabilities required to exchange clinical, insurance, and prior authorization data between the hospital EHR, MediConnect platform, and payer systems.

Scope note: These requirements focus on functional behavior only. Non-functional requirements such as security, performance, and availability should be captured separately.

1. User Access and Role Management

- The system shall support role-based access for providers, care coordinators, billing staff, and administrators.
- The system shall allow administrators to create, update, disable, and audit user accounts.
- The system shall restrict access to patient records and authorization workflows according to the user role and assigned organization.

2. Patient and Case Intake

- The system shall capture patient demographic data required for prior authorization submission.
- The system shall capture payer, plan, policy, and member information for each authorization case.
- The system shall allow users to create a new prior authorization case manually or from an incoming EHR referral.
- The system shall preserve a case identifier that links the request across all downstream transactions.

3. Clinical Data Acquisition

- The system shall receive patient clinical data from the hospital EHR through standardized integration interfaces.
- The system shall ingest diagnosis codes, procedure codes, medications, allergies, notes, and supporting documentation when available.
- The system shall validate that required clinical fields are present before the case is submitted to the payer workflow.

4. HL7 Integration

- The system shall accept HL7 messages from the hospital EHR and associate them with the correct patient or case record.
- The system shall validate HL7 message structure and reject malformed messages with an actionable error message.
- The system shall log inbound and outbound HL7 message activity for traceability.

5. HL7 to FHIR Transformation

- The system shall transform supported HL7 data elements into FHIR resources for downstream processing.
- The system shall map patient, encounter, condition, observation, and document data into the appropriate FHIR structures.
- The system shall flag unmapped or incomplete data elements for review rather than silently dropping them.

6. FHIR API Services

- The system shall expose FHIR-based APIs for exchanging patient and authorization data.
- The system shall support create, read, update, and query operations for authorization-related FHIR resources as defined by the solution design.
- The system shall return standardized API responses that indicate success, validation errors, and downstream processing states.

7. Prior Authorization Workflow

- The system shall create a prior authorization request from the available clinical and insurance data.
- The system shall route the request to the correct payer based on plan and payer rules.
- The system shall support request states including Draft, Submitted, Pending Review, Approved, Denied, and More Information Required.
- The system shall allow users to add supplemental documents and resend the request when additional information is requested.

8. Payer Response Handling

- The system shall receive payer decisions and store them against the associated authorization case.
- The system shall support payer responses for approval, denial, and request for additional information.
- The system shall capture payer reasons, remarks, and next-step instructions where provided.

9. EHR Status Updates and Notifications

- The system shall send authorization status updates back to the hospital EHR.
- The system shall notify relevant users when a case changes status or requires action.
- The system shall support notifications for approval, denial, missing information, and transmission failure events.

10. Audit Trail and Tracking

- The system shall maintain an audit trail for case creation, updates, submission, response handling, and user actions.
- The system shall store timestamps, user identifiers, message identifiers, and status transitions for each transaction.
- The system shall allow authorized users to view the history of a case from intake through final decision.

11. Reporting and Monitoring

- The system shall provide operational reporting for case volume, turnaround time, approval rate, denial rate, and pending cases.
- The system shall identify stalled cases and cases missing required information.
- The system shall provide exportable reports for operational review and management analysis.

Requirement Summary Matrix

ID	Requirement Area	Description
FR-01	User Access And Role Management	See the corresponding functional requirement section above.
FR-02	Patient And Case Intake	See the corresponding functional requirement section above.
FR-03	Clinical Data Acquisition	See the corresponding functional requirement section above.
FR-04	HI7 Integration	See the corresponding functional requirement section above.
FR-05	HI7 To Fhir Transformation	See the corresponding functional requirement section above.
FR-06	Fhir Api Services	See the corresponding functional requirement section above.
FR-07	Prior Authorization Workflow	See the corresponding functional requirement section above.
FR-08	Payer Response Handling	See the corresponding functional requirement section above.
FR-09	Ehr Status Updates And Notifications	See the corresponding functional requirement section above.
FR-10	Audit Trail And Tracking	See the corresponding functional requirement section above.
FR-11	Reporting And Monitoring	See the corresponding functional requirement section above.